Engine Model(s): PT6A

79-20-04

OIL-TO-FUEL HEATER CHECK VALVE - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC30518	Puller	
PWC50986	Puller, Check Valve	Alternate to PWC30518

4. Fixtures, Equipment and Supplier Tools

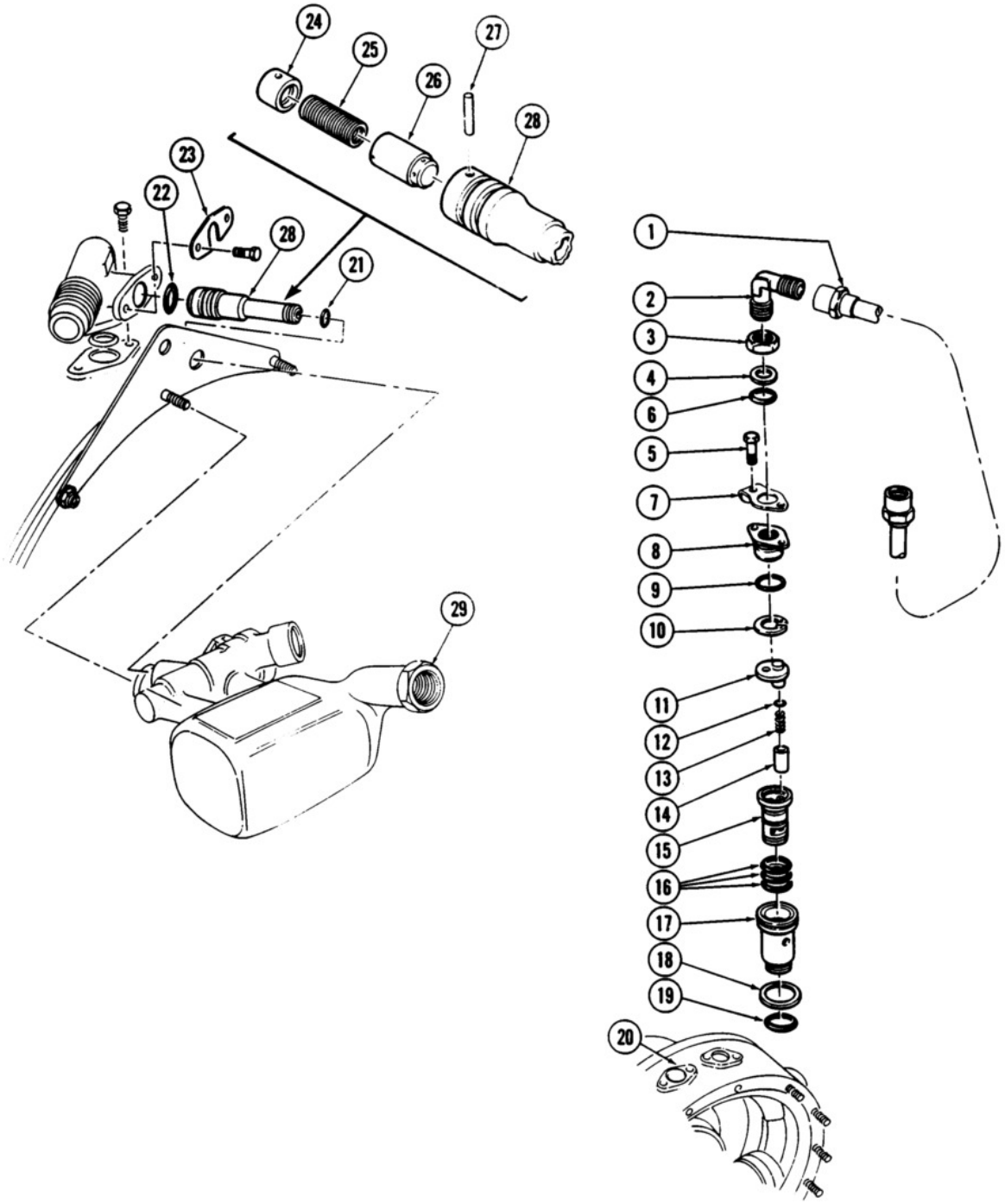
Not Applicable

5. Removal/Installation

A. Removal of Oil-to-Fuel Heater Check Valve (Ref. Fig. 201)

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Figure 201 Removal/Installation of Oil-to-Fuel Heater Check Valve



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1. Oil Inlet Tube
2. Elbow
3. Locknut
4. Back-up Ring
5. Bolt
6. Preformed Packing
7. Fireseal Support Bracket
8. Adapter
9. Preformed Packing
10. Retaining Ring
11. Spring Seat
12. Preformed Packing
13. Spring
14. Check Valve (Minimum Pressurizing)
15. Valve Sleeve
16. Preformed Packings
17. Valve Housing
18. Preformed Packing
19. Preformed Packing
20. Compressor Inlet Case
21. Preformed Packing
22. Preformed Packing
23. Retaining Plate
24. Spring Seat
25. Spring
26. Check Valve (Oil Return)
27. Spring Pin
28. Valve Housing
29. Oil-to-Fuel Heater

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(1) Remove oil inlet tube (1) and oil-to-fuel heater (Ref. 73-10-01).

(2) Remove two bolts and remove retaining plate (23) from adapter elbow at top of compressor inlet case.

(3) Withdraw check valve housing (28) from adapter elbow and remove preformed packings (21 and 22) from housing.

(4) Disassemble oil return check valve assembly:

(a) Remove spring pin (27) from valve housing (28).

(b) Remove spring seat (24), spring (25) and check valve (26) from housing (28).

(5) Remove elbow (2) and locknut (3) from adapter (8) at 2 o'clock position on inlet case. Discard preformed packing (6) and back-up ring (4).

CAUTION: TAPE NUT TO FIRESEAL TO MAKE SURE THAT FRONT LOCKNUT SETTING IS NOT DISTURBED. THIS WILL AVOID READJUSTMENT OF REVERSING LINKAGE.

(6) Release rear locknut securing rear swivel joint and rear fireseal support bracket (7) to fireseal mount ring.

(7) Remove two bolts (5) and pivot support bracket outwards from engine.

(8) Remove adapter (8). Discard preformed packing (9).

(9) Remove retaining ring (10) and remove spring seat (11), spring (13) and check valve (14) from valve sleeve (15). Discard preformed packing (12).

(10) Use pullers (PWC30518) or (PWC50986), remove valve housing (17) from compressor inlet case.

(11) Remove sleeve (15) from valve housing (17), remove preformed packings (16) from valve sleeve, and preformed packings (18 and 19) from valve housing.

B. Installation of Oil-to-Fuel Heater Check Valve (Ref. Fig. 201)

(1) Assemble and install minimum pressurizing check valve:

(a) Install preformed packing (12) on spring seat (11).

(b) Insert check valve (14), spring (13) and spring seat (11) into valve sleeve (15). Secure assembly with retaining ring (10).

(c) Install three preformed packings (16) on valve sleeve (15) and insert sleeve in valve housing (17).

(d) Install preformed packings (18 and 19) on valve housing (17).

(e) Install complete assembly of check valve in ported well on right-hand side of inlet case (20). Ensure assembly is fully bottomed.

(f) Install preformed packing (9) on adapter (8). Locate adapter in ported well and align mounting holes.

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(g) Pivot rear fireseal support bracket (7) inward over adapter (8). Secure adapter and bracket with two bolts (5). Tighten bolts, torque 36 to 40 lb.in., and lockwire.

(h) Install elbow (2), locknut (3), back-up ring (4) and preformed packing (6) on adapter (8) (Ref. 70-00-00, REMOVAL/INSTALLATION). Do not final torque locknut (3) until angle of elbow has been established (Ref. Subpara. (3)).

(2) Assemble and install oil return check valve:

(a) Insert check valve (26), spring (25) and spring seat (24) into valve housing (28). Align holes in seat and housing, and secure assembly with spring pin (27).

(b) Install preformed packing (22) on housing (28) and insert complete check valve assembly into elbow adapter at top of inlet case.

(c) Secure valve assembly in elbow adapter with retaining plate (23) and two bolts. Tighten bolts, torque 32 to 36 lb.in., and lockwire.

(3) Install preformed packing (21) on valve housing (28), and install oil-to-fuel heater (29) and oil inlet tube (1) (Ref. 73-10-01). At installation of tube (1), final torque locknut (3) on elbow (2) 38 to 42 lb.in., and lockwire.

(4) Replenish oil tank (Ref. 72-00-00, SERVICING).

(5) Check for oil leaks at engine test run (Ref. 71-00-00).

6. Cleaning/Painting

A. Clean Check Valve

(1) Clean all components of the minimum pressurizing check valve with petroleum solvent (PWC11-027) or (PWC11-031)Make sure the check valve and interior surface of the respective housings are completely clean.

7. Inspection/Check

A. Inspect Check Valve

(1) Examine oil return check valve housing for general condition. Raised metal may be removed by light buffing, especially in areas of mating ends.

(2) Examine the elbow of the minimum pressurizing check valve for the condition of the threads. Light damage may be cleaned up be chasing.

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